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Care Creations™

Nephydrat™

The Bio-Inspired Hydro-Energizer

by Beauty Creations
another Care Creations™ product group Inspired by Life

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SUMMARY FILE

Nephydrat	**BC10061**
Origin - Description	Natural extract of rambutan pericarps (peels), titrated in polyphenols, from a sustainable sourcing program
Regulatory data	
INCI	Glycerin (and) Water (and) Nephelium Lappaceum Peel Extract
China	Each component is listed in “Inventory of Existing Cosmetic Ingredient in China” (IECIC 2015)
CAS#	56-81-5; 7732-18-5; 93165-68-5 / 999999-99-4
EINECS#	200-289-5; 231-791-2; 296-955-8 / 310-127-6
Appearance	Liquid
Preservative	None
Natural labels	Raw material conforms to COSMOS standard of natural and organic cosmetics
Naturalness content (ISO 16128)	100% from natural origin
Cosmetic use	
Properties	Strengthens *stratum corneum* Energizes cells Maintains cellular hydration
Applications	Skin hydration Skin energizer Natural cosmetics Eco-responsible face care formulations
Formulation data	
Concentration of use	2%
Incorporation method	Nephydrat is incorporated during the final process at a temperature below 30°C
Optimal temperature of use	15-30°C
Optimal pH	3-7.5
Patent family	Application filed



SUSTAINABLE UPCYCLING OF RAMBUTAN IN THE SERVICE OF BEAUTY

Responsible living is a daily challenge. Women and men want to give their skin special care to counter environmental impacts, but not to the detriment of the planet. Consumers are now searching for more ethical cosmetic products, which are as healthy for their skin as they are for the earth. Consequently, the cosmetic industry has taken up the trends of the demand of modern consumers and their growing environmental awareness to continue developing sustainable solutions. This industry expects more responsible cosmetic ingredients from suppliers who can ensure a reliable supply chain and social commitment, and who will not compromise on the level of effectiveness. Therefore, we have embedded sustainability in our corporate purpose: “We create chemistry for a sustainable future.”

This is the starting point of our work. Capitalizing on our experience acquired with a previous Argan program, we have discovered a plant and built a new specific supply chain in cooperation with a long-standing local partner in Vietnam. We have identified rambutan, also called *Nephelium lappaceum*, a well-known tree in Asia, that grows in the humid tropics. Out of the fruit itself, we have found that other by-products of the rambutan plant possess interesting properties for our skin’s health and beauty (Figure 1).

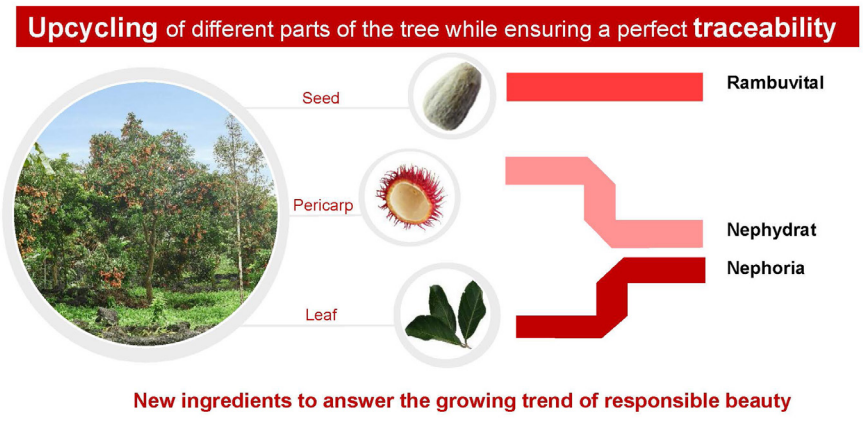


Figure 1 - Upcycled active ingredients from rambutan tree.

Nephydrat, our unique active ingredient extracted from the protective envelope of sustainably sourced rambutan, rescues dry and weakened skin with a 3-step strategy: to reinforce the *stratum corneum*, to boost cell energy and to maintain cellular hydration.

ACTING BOTH ON THE BRICKS AND THE CEMENT OF THE SKIN BARRIER FOR A REINFORCED EPIDERMIS

The skin's most important role is to form an effective barrier between the "inside" and the "outside" of the organism. The epidermis, and more particularly the *stratum corneum* (SC), is the physical barrier of the skin. It is mainly involved in the regulation of water release from the organism and into the atmosphere (Proksch *et al.*, 2008).

The formation of the epidermis is a slow and coordinated process of cell proliferation, differentiation and death (Candi *et al.*, 2005) (Figure 2). When the basal keratinocytes end the proliferation phase (*stratum basale* layer), they start to differentiate (*stratum spinosum* layer) and migrate up to the skin surface (*stratum corneum* layer) to become corneocytes, which are dead cells presenting a rigid and insoluble double envelope: the cornified envelope.

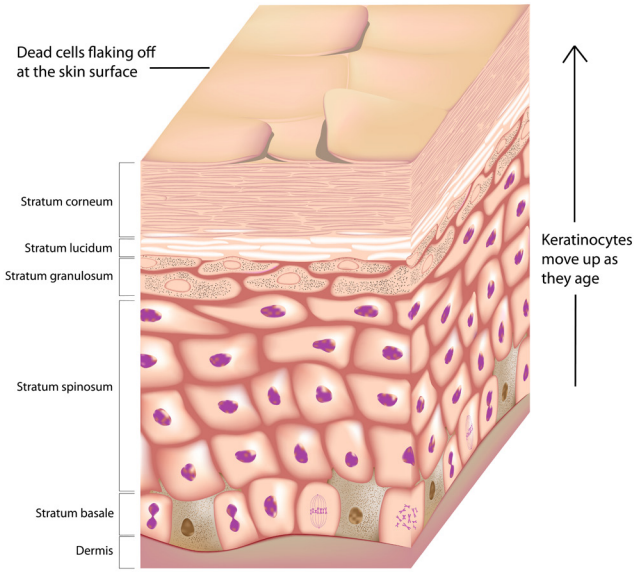


Figure 2 - Anatomy of the epidermis.

Schematically, the SC has a typical "brick and mortar (cement)" type structure (Figure 3). The corneocyte "bricks" occupy most of the volume of the SC wall and are surrounded by a lipidic cement (Moore and Rawlings, 2017). In human SC, up to around 25 corneocyte layers are present (Van Smeden *et al.*, 2016).

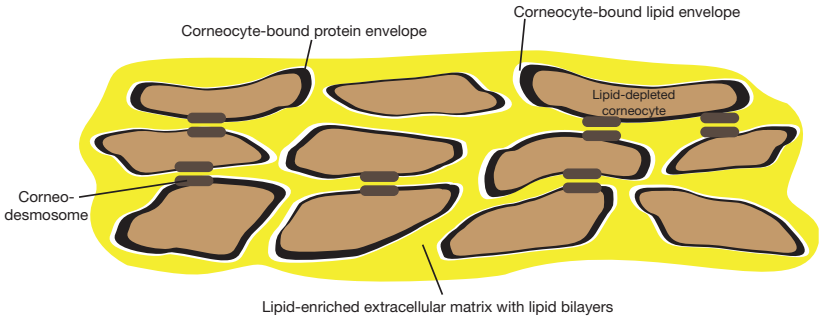


Figure 3 - Brick and cement (mortar) structure of the *stratum corneum*. Each corneocyte is surrounded by an inner protein envelope and an outer lipid envelope (according to Proksch *et al.*, 2008).

As shown in Figure 3, **the cornified envelope** consists of two parts: a protein envelope and a lipid envelope (Nemes and Steinert, 1999). This protein-lipid envelope contributes to the mechanical resistance and hydrophobicity of the *stratum corneum* (Guneri *et al.*, 2018).

- **The protein envelope** is thought to contribute to the biomechanical properties of the cell envelope, resulting from the cross-linking of involucrin (IVL), loricrin, filaggrin and some small proline rich proteins by transglutaminases (TGs). This reticulated protein envelope is resistant to most common proteolytic enzymes (Proksch *et al.*, 2008; Candi *et al.*, 2005). Even if its complete role is not precisely known, hornerin was also shown to be a substrate of TG (particularly transglutaminase 3) and consequently is considered to be a component of the cornified envelope (Henry *et al.*, 2011; Hsu *et al.*, 2017).
- **The lipid envelope** is located on the external part of the corneocyte and is covalently attached to the protein envelope. It consists of ω-hydroxy ceramides, mainly long chain ceramides (with number of carbons > 30) covalently linked by ester bonds to cell envelope proteins, most abundantly to involucrin (Nemes *et al.*, 1999) (Figure 4). This lipid envelope creates a hydrophobic coating which surrounds the corneocytes and contributes to the organization of intercellular lipids, the cement (Marekov and Steinert, 1998; Elias *et al.*, 2014).

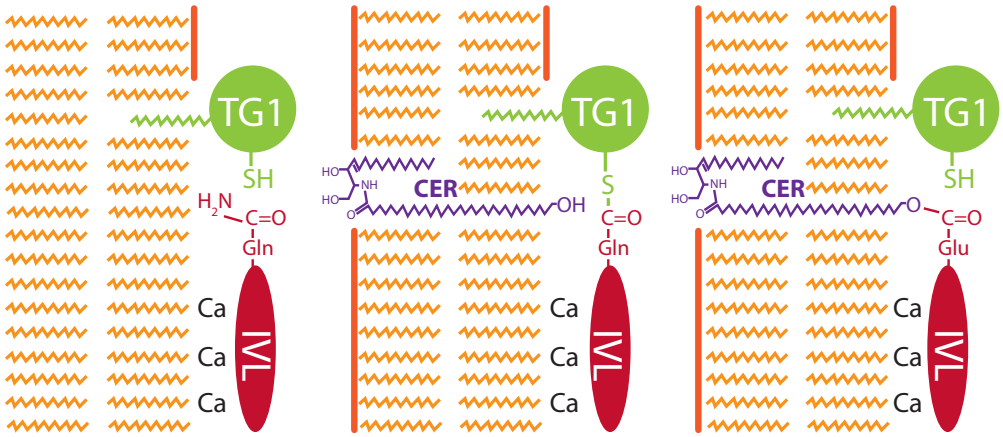


Figure 4 - Model for the chemical reaction between involucrin and ω-hydroxyceramide (according to Nemes *et al.*, 1999). Cross-linking reaction catalyzed by transglutaminase-1 (TG1) between involucrin (IVL) and ω-hydroxyceramide (CER).

The cornified envelope contributes to the structure and hydrophobicity of the *stratum corneum* for optimal barrier function and a well-hydrated skin.

The corneocytes are imbedded in a lipophilic extracellular medium, the **lipidic cement**, structured into a unique lamellar organization composed of ceramides (50%), cholesterol (25%) and fatty acids (10-20%) (Wertz, 2000; Pappas, 2009).

- **Ceramides** are crucial to the structure and maintaining the integrity of the skin barrier (Rabionet *et al.*, 2014). As the major lipid constituent of the cement, ceramides play an essential role in the water permeability barrier function of the skin. In conjunction with the other *stratum corneum* lipids, they form organized structures (Codech *et al.*, 2003). In addition, ceramides and their metabolites have been recognized as signaling lipids that modulate cellular function such as proliferation, differentiation, and apoptosis in epidermal keratinocytes (Uchida, 2014).

  Free long chain ceramides (>40 carbons) have been reported to be essential for good barrier function (Moore and Rawlings, 2017). In addition, long-chain ω-hydroxyceramides are covalently linked to the outer surface of the corneocytes by transglutaminase-1, most abundantly to involucrin (Figure 4) (Nemes *et al.*, 1999; Marekov and Steinert, 1998). The chains of the ceramides are then ordered to form a water barrier around each corneocyte, for optimal barrier function (Figure 5).

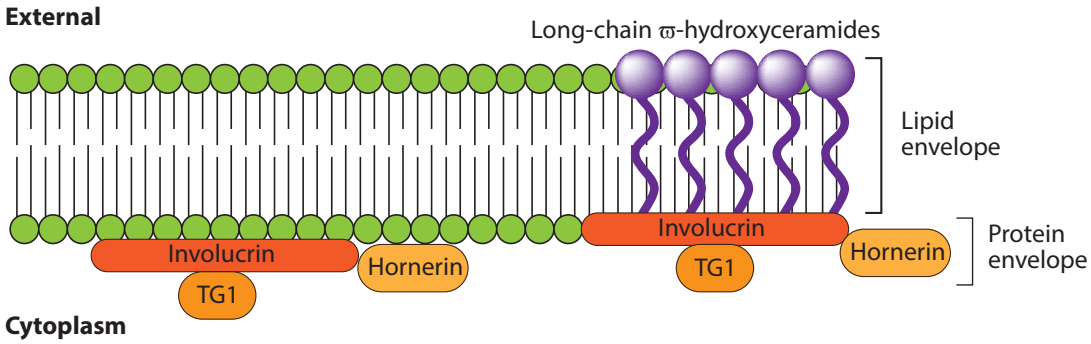


Figure 5 - Schematic representation of the organization of the cornified envelope.

- **Cholesterol** plays an important role in skin barrier homeostasis by stabilizing SC lipid organization and inducing conformational ordering (Bouwstra *et al.*, 1999; Sochorova *et al.*, 2019). In addition, when first sulfated by an enzyme (cholesterol sulfotransferase) and then desulfated in the outer epidermis (Figure 6), cholesterol and cholesterol sulfate form a "cholesterol sulfate cycle" that can regulate epidermal differentiation, barrier function and desquamation (Elias *et al.*, 2014).
- **Fatty acids** are required for barrier homeostasis (Mao-Qiang *et al.*, 1993). They are biosynthesized from phospholipids initially present in keratinocytes, which are then degraded by phospholipases as the cells enter the *stratum corneum*, to become free fatty acids (Figure 6). These free fatty acids contribute to *stratum corneum* acidification, integrity and cohesion (Fluhr *et al.*, 2001; Feingold and Elias, 2014).

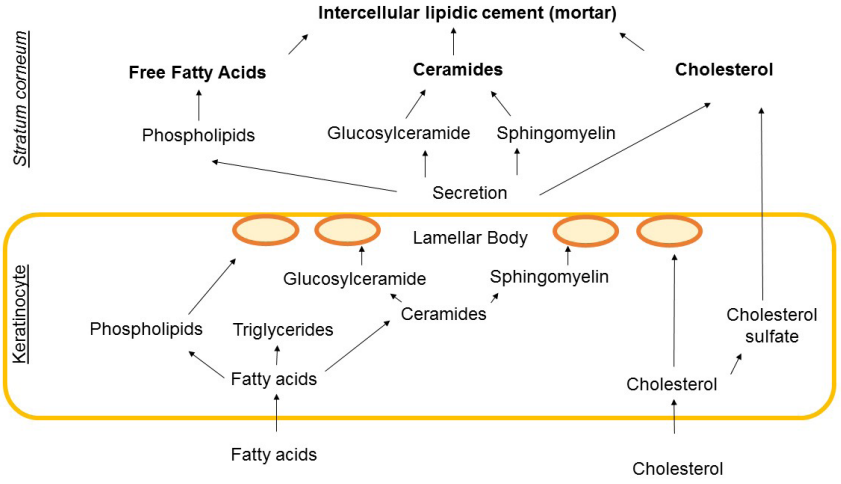


Figure 6 - Schematic representation of lipidic cement synthesis (according to Feingold and Elias, 2014).

The intercellular lipids contribute to the structure and hydrophobicity of the *stratum corneum* for optimal barrier function and well-hydrated skin.

A lack of lipidic cement or bad quality of the cornified envelope (bricks) result in dry skin. Improving both the bricks and the cement is a target for moisturizing ingredients.

RE-ENERGIZING SKIN CELLS AND PROTECTING MITOCHONDRIA FOR OPTIMAL FUNCTION

All cell syntheses (proteins, lipids, enzymes…) and cell metabolisms (proliferation, differentiation…) require energy. Adenosine triphosphate, or ATP, is the energy currency of life. ATP is produced by mitochondria as part of the Krebs cycle (Figure 7).

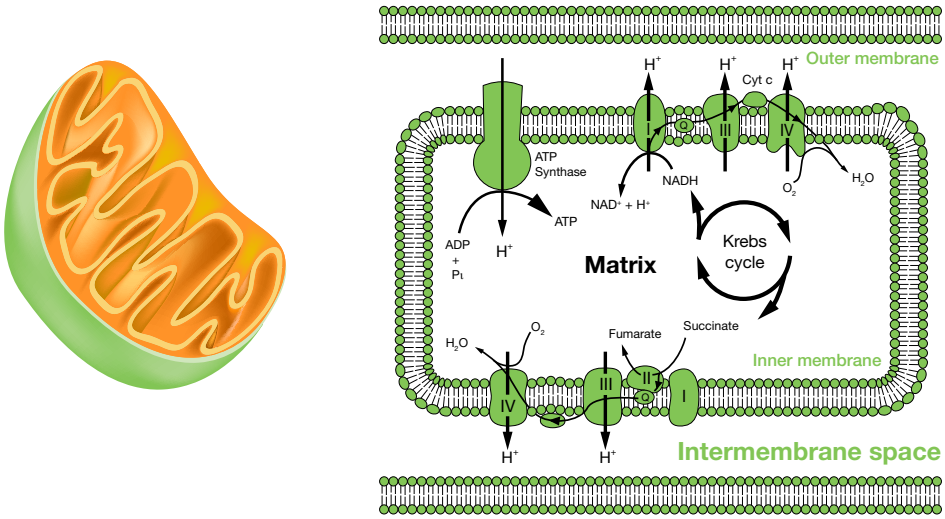


Figure 7 - Schematic representation of the Krebs cycle.

Mitochondria are ubiquitous and unique cellular organelles having their own genome. They are composed of two membranes with an inner membrane involving the molecules of the respiratory chain and delimiting the central matrix. The main functions of mitochondria are the production of energy (ATP synthesis) by the respiratory chain but also lipid oxidation, calcium storage, amino acid metabolism, hormonal signaling and control of apoptosis (Kudryavtseva *et al.,* 2016). As mitochondria are involved in many essential processes for cell survival, they are key actors in skin physiology. They participate in epidermal differentiation and in stem cells maintenance (Feichtinger *et al.*, 2014). In addition, mitochondria participate in synthesis of key components of the human epidermis such as steroids and glucocorticoids (Slominski *et al.*, 2017).

Mitochondrial membrane potential (MiMP) is a key indicator of the “health” of mitochondria. A high MiMP level is critical for maintaining the physiological function of the respiratory chain to generate ATP and consequently plays a key role in mitochondrial homeostasis. MiMP is also a driving force for the transport of ions and proteins which are necessary for healthy mitochondrial function (Zorova *et al.*, 2018). MiMP is quite stable for normal cell function (Zamzami *et al.*, 1995) but a significant loss is observed when the cells are depleted of energy with subsequent death (Joshi and Bakowska, 2011).

Mitochondria can form physically interconnected network (fusion step) that may represent an efficient system of delivering energy with an increase of ATP production (Hoitzing *et al.*, 2015; Skulachev, 2001). During apoptosis, mitochondrial fragmentation with a disintegration of this tubular interconnected mitochondrial network is observed (Karbowski and Youle, 2003).

Mitochondria are the “powerhouses” which synthesize ATP inside the cells. Boosting their ATP production and protecting them against stresses is a target for maintaining optimal cell functions in general, and optimal epidermal homeostasis in particular for well-hydrated skin.

STRENGTHENING THE CELL PROTECTION SYSTEM AGAINST ENVIRONMENTAL CHANGES: TAURINE AS A KEY OSMOREGULATOR

As the uppermost layer of the skin, the epidermis is strongly involved in the regulation of spontaneous water release from the organism to the atmosphere. Under baseline conditions, water loss from the *stratum granulosum* layer (Figure 2) is compensated by replenishment via the lower layer of the epidermis. Osmolytes such as taurine (2-aminomethane sulfonic acid, Figure 8) are required to maintain a constant cell volume, which is crucial for the maintenance of optimal cell functions, and intracellular water content in the outer part of the epidermis (Anderheggen *et al.*, 2006; Janeke *et al.*, 2003).

Endogenous taurine synthesis occurs mainly in the liver and brain (Worden and Stipanuk, 1985) but the main source is provided by animal-derived foods, particularly meats and seafood (Lourenço and Camilo, 2002). Taurine has several biological functions such as calcium level modulation, synaptic activity neuromodulation and membrane stabilization. It is also an inhibitiory neurotransmitter. In the skin, its function as osmoregulator is crucial for water regulation (Grafe *et al.*, 2004; Ripps and Shen, 2012). The highest taurine content is found in the *stratum granulosum* and upper *stratum spinosum* layers (Figure 2) where the cells are the most vulnerable to environmental fluctuations in humidity (Anderheggen *et al.*, 2006). Exposed to a dry environment, taurine is taken up from the extracellular compartment via the Na^+-dependent taurine transporter (TauT) and released inside the cell. The accumulation of taurine induces an increase of intracellular osmolarity and allows the re-diffusion of water into the cell to restore normal cell volume, water and electrolyte concentration (Janeke *et al.*, 2003; Anderheggen *et al.*, 2006).

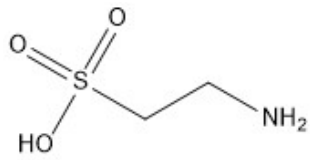


Figure 8 - Taurine molecule.

In the epidermis, taurine is a key osmoregulator (an equalizer of cell water content), required for keratinocyte hydration, and which enters the cell via its transporter, TauT.

Nephydrat

A sustainable ingredient for a moisturized skin



Description of the plant

Family: Sapindaceae

Species: *Nephelium lappaceum* L.

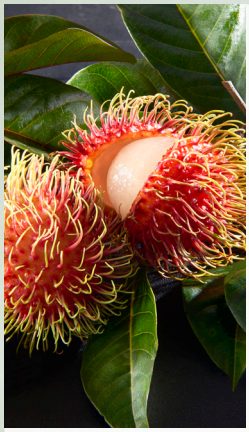
Common name: the name "rambutan" comes from the Malay word “rambut” meaning "hair", a reference to the numerous hairy protuberances on the fruit. In Vietnam, rambutan is called chôm chôm (meaning "messy hair").

Description: *Nephelium lappaceum* is an evergreen tree that reaches 12 to 20 meters high in nature and 4 to 12 meters when cultivated clonaly. Its main trunk has an open crown of large branches. Its bark is slightly rough and greyish or red. The hairy fruit is mainly red in color, but may also can be also yellow, orange or pink. The tree has a life spend of up to 100 years.

Distribution: its origin, possibly the Malay Archipelago, is undeterminable because wild growth from cultivation has blurred the contours of its primary distribution. It was then disseminated to the South Asian countries of Vietnam, Thailand and China. Nowadays, the extend of rambutan cultivation ranges in South China to the Philippines and is found in tropical areas of South America.

Habitat: Rambutan trees grow in humid tropics, especially in elevation of up to 700 meters, with temperatures comprised between 22 and 30°C, in areas exposed to the sun and subjected to heavy rains (between 2000 and 5000 mm/year).

Bibliography available upon request.



Nephelium lappaceum: traditional and modern uses

Rambutan, whose fruit is edible, is already known for its significant benefits to health and is widely used in Asia (Hernandez-Hernandez *et al.,* 2019).

The fruits have long been used to quell dysentery while the peels (pericarps) contain a variety of beneficial compounds that demonstrate antioxidant and anti-cancer properties. The leaves are used to treat headaches and the bark are used for tongue diseases.

The sweet-tasting fruit is consumed fresh or canned. It is appreciated for its refreshing flavor and exotic appearance. Naturally highly rich in iron, vitamin C, fibers and antioxidants, rambutan is also reported to have some benefits similar to superfruits. For example, it is reported to diminish fat, to act against free radicals, to boost immunity, to prevent diseases, and so on.



The responsible rambutan program: upcycling for greater social impact and sustainable supply chain

Origin: Vietnam

Part used: Pericarp (peel), for Nephydrat
Other parts, leaves and seeds, are also used for the other active ingredients, Nephoria and Rambuvital (scientific brochures also available).

Variety: Java

Supply and cultivation: in 2015, a specific supply chain was set up in cooperation with our long-term supplier in Vietnam. Rambutan trees are cultivated in organic-certified gardens in Southern Vietnam. Our supplier and the network of farmers involved in production promote sustainable agricultural practices, respecting the local ecosystem and improving working conditions including safety higher incomes for workers and specific on-the-job training.

Harvesting and processing: the main harvest of rambutan occurs from July to September. A tree produces up to 200 kg of fruit per year. Fruit is gathered in clusters at maturity and leaves are removed for later use. The premium quality fruit is then processed according to a specific technic know-how.

Seeds, pericarps and leaves are then processed to produce active ingredients for use in cosmetics.

Processing and Composition



Nephydrat is obtained through a nature-friendly extraction process using water as sole solvent. Following a first step of aqueous extraction of the grinded pericarps (peels), the unextracted insoluble matter is removed. Glycerin is then added to obtain Nephydrat as a standardized preservative-free liquid (Glycerin 80%; *Nephelium Lappaceum* Peel Extract 0.3-0.6%, sqf 100% water) titrated in polyphenols.

**No involvement of animal genes or animal-derived substances; unlikely cross-contamination from animal substances; no animal testing has been carried out by or on behalf of BASF on the ingredients of the product after 11th March 2009 and/or after the extended deadline 11th March 2013 for the purposes of the Cosmetic Regulation (EU) 1223/2009 (statement available upon request).

Nephelium lappaceum peel (pericarp) extract, i.e. the plant matter, contains mostly polyphenols and carbohydrates (Figure 9). Among the polyphenols, two other main compounds are geraniin and corilagin (Figure 10A, 10B and 10C) belonging to the ellagitannin family.

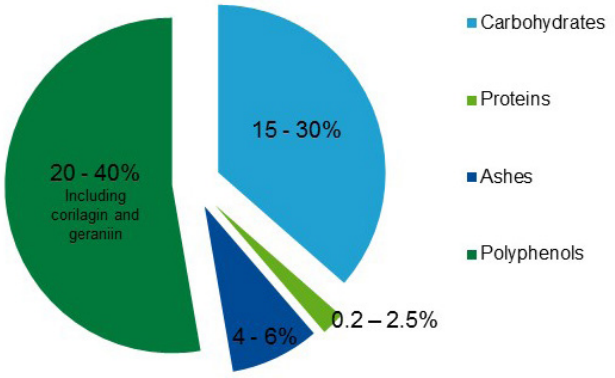


Figure 9 - Typical phytochemistry composition of a *Nephelium lappaceum* peel (pericarp) extract matter (based on 3 representative batches) in dry weight percentages.

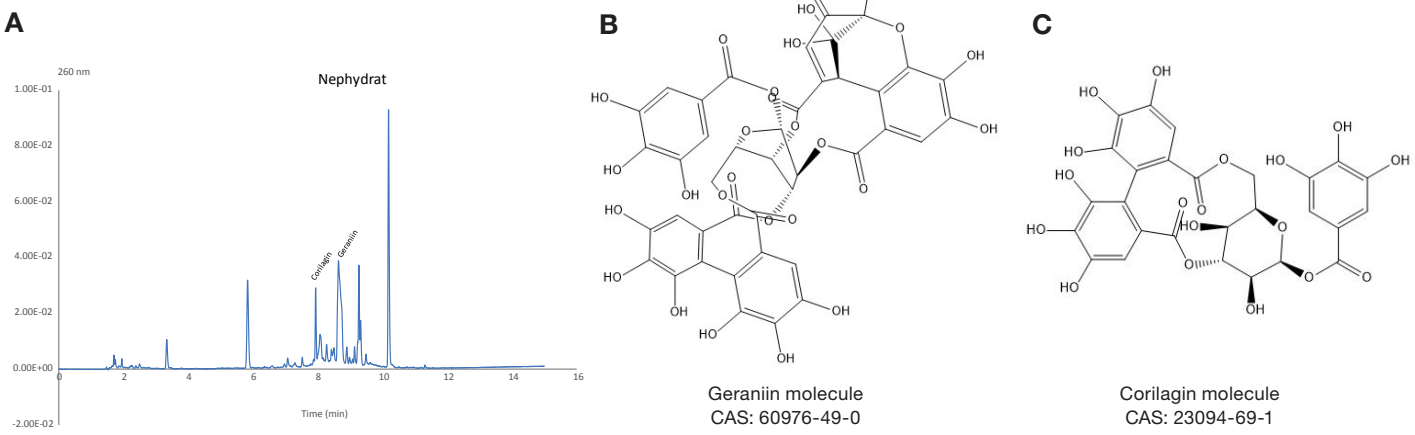


Figure 10 - **A** - Example of HPLC-PDA typical chromatogram profile (260 nm). **B** - Molecular structure of geraniin. **C** - Molecular structure of corilagin.

Safety / tolerability of the product

Nephydrat was tested to ensure its safety under the recommended conditions of use. Nephydrat does not irritate the eyes or skin and no indication of skin sensitization was observed

Description

Nephydrat is a natural standardized preservative-free extract of rambutan peels (pericarps), titrated in polyphenols, from a sustainable sourcing program.

INCI name: Glycerin (and) Water (and) Nephelium Lappaceum Peel Extract

Naturalness content (according to ISO 16128): 100% from natural origin

Dose of use: 2%

China IECIC listed

Approved by ECOCERT RAW MATERIAL COSMOS APPROVED

Vegan Friendly **



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Example: illustration of a commercial sample of Nephydrat, with an emulsion and hydrogel containing 2%.

**No involvement of animal genes or animal-derived substances; unlikely cross-contamination from animal substances; no animal testing has been carried out by or on behalf of BASF on the ingredients of the product after 11th March 2009 and/or after the extended deadline 11th March 2013 for the purposes of the Cosmetic Regulation (EU) 1223/2009 (statement available upon request).



DEMONSTRATED EFFICACY

Nephydrat has been proven to increase skin hydration through *in vitro* and *in vivo* tests

Proven *in vitro* performance

Nephydrat has proven its efficacy *in vitro* to optimize skin hydration through a 3-step strategy (Figure 11).

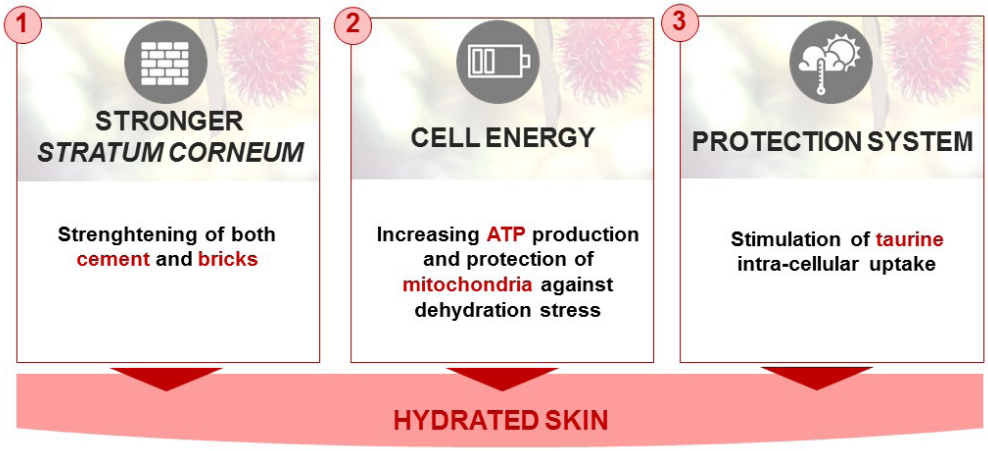


Figure 11 - Summary of the 3-step strategy of Nephydrat for optimized skin hydration.

Proven *in vivo* performance

In vivo, Nephydrat has been proven to increase skin hydration in a one-month placebo-controlled study on the face.

In a second approach, the benefits on the skin of Nephydrat formulations were confirmed by 3 independent panel assessment studies in Europe, Asia and North America.

EFFICACY

REINFORCING THE CEMENT AND THE BRICKS FOR A STRONGER *STRATUM CORNEUM*

Stimulation of the lipid components of the cement

in vitro

OBJECTIVE

In the *stratum corneum*, a layer of lipids fills the spaces between the cells, like mortar or cement. Combined with corneocytes (bricks), these lipids form the natural skin barrier and protect the skin from water loss.
The aim of this study was to evaluate the effect of Nephydrat on lipid synthesis, using a 3D model of reconstructed epidermis.

RESULTS & DISCUSSION

Figure 12 shows the effects of Nephydrat tested at 0.1% on phospholipid and ceramide stimulation. Nephydrat significantly increased the phospholipid synthesis by 15%, which is the first step in the biosynthesis of the fatty acid components of the lipidic cement.
Total ceramides, including long chain ceramides (>40 carbons), which are essential in lipid organization and barrier function (Moore and Rawlings, 2017), increased respectively by 20% and 12%.

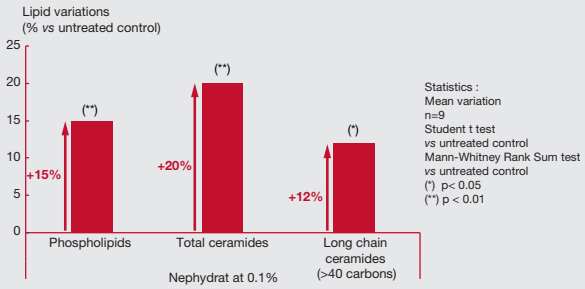


Figure 12 - Effect of Nephydrat on lipid synthesis.

As shown in our 3D reconstructed epidermis, Nephydrat stimulates both phospholipid and ceramide synthesis, including long-chain ceramides, to reinforce the *stratum corneum* and its barrier function against skin dehydration.

Optimization of the brick envelopes

in vitro

OBJECTIVE

In the *stratum corneum*, transglutaminase-1, hornerin and involucrin proteins are components of corneocyte bricks involved in skin protection from water loss.
The aim of this study was to evaluate *in vitro* the effect of Nephydrat on the synthesis of these proteins by human keratinocytes.

RESULTS & DISCUSSION

Figure 13 shows the dose-dependent effect of Nephydrat on these hydration targets. Nephydrat significantly increased the synthesis of transglutaminase-1 (TG1) (Figure 13A) by 81% at 0.05%. In addition, involucrin synthesis was significantly stimulated up to 79% at 0.3% (Figure 13B). Hornerin, another component of the cornified cell envelopes was significantly stimulated up to 81% at 0.05% (Figure 13C). By stimulating both involucrin and hornerin synthesis and their cross-linking enzyme, transglutaminase-1, Nephydrat reinforces the corneocyte protein envelope.
As outlined previously, long-chain ceramide lipids are covalently bonded to the cornified cell envelope through an enzymatic reaction involving transglutaminase-1 and involucrin (Figure 4). As Nephydrat stimulates both transglutaminase-1, involucrin and long chain-ceramides (Figure 13A, 13B and Figure 12), it should act on the assembly of the lipid envelope for a better barrier function.

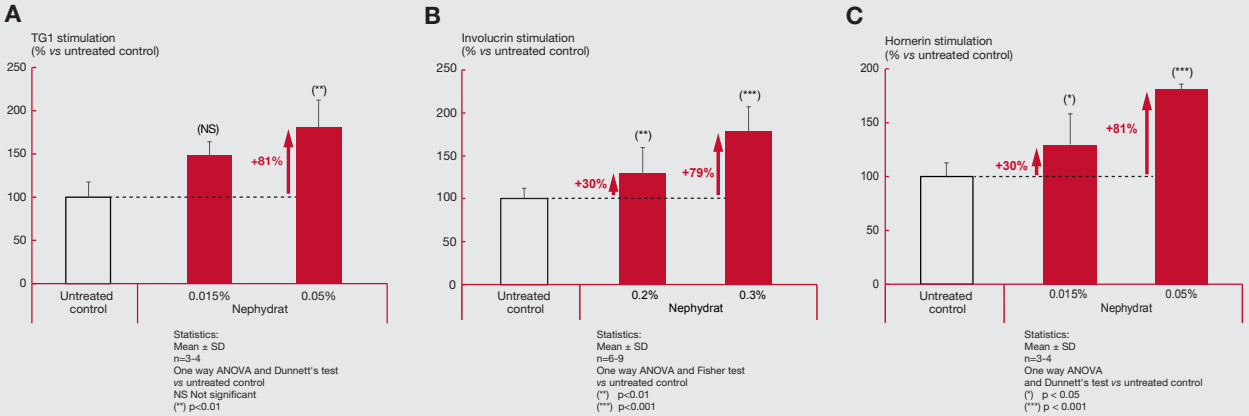


Figure 13 - Effect of Nephydrat on the protein expression of hydration targets. **A** - Transglutaminase-1 synthesis. **B** - Involucrin synthesis. **C** - Hornerin synthesis.

Nephydrat stimulates both transglutaminase-1, involucrin, hornerin, and long-chain ceramides to form stronger and more hydrophobic "bricks" (corneocytes), leading to a reinforced *stratum corneum*.

CONCLUSION

Nephydrat reinforces *in vitro* both the bricks and the cement (mortar) of the skin barrier, for optimal barrier function.

MATERIALS&METHODS

Reconstructed epidermis

Cell culture

Reconstructed epidermises (n=9 for each condition) were prepared as follows. 30000 cells/cm² of 18 years old fibroblasts were seeded under a 0.47 cm² Nunc Carrier Plateform Cell Culture. Two days later, 250 000 cells/cm² of keratinocytes from a 33 years old donor were seeded on the other side of the Nunc membrane. DMEM-Ham medium with 10% bovine calf serum was changed daily for 4 days. Then, each model was put at the air-liquid interface and cultured for 7 days. The medium composed of DMEM-Ham with 5% fetal calf serum (FCS) was changed daily for 4 days and DMEM-Ham modified with 1% of fetal calf serum was used for 3 additional days. *Nephelium lappaceum* peel extract at a dose equivalent to Nephydrat 0.1% was added into the culture medium from day 1 of the air-liquid interface.

At the end of the culture, reconstructed epidermises were washed in a phosphate buffered saline (PBS) solution for 2 minutes and then frozen at -80°C in cryovials.

Lipid analysis

Reconstructed epidermises were cryo-broken into an aqueous solution and then extracted in an organic solvent blend overnight. Phospholipids were then analyzed by liquid chromatography (LC) coupled with UV and fluorescence detection. For ceramide analysis, several successive extractions were performed for purification purposes. Analysis and quantification of total and long-chain ceramides was done with liquid chromatography coupled with a mass spectrometry method (LC/MS).

Results and statistics

The results are expressed as the mean percentage of change versus the untreated control. The statistical comparison of the variation of lipid classes between treated and untreated epidermises was done using Student's t test or Mann-Whitney Rank sum test, following Sigmaplot® software recommendations (Systat Software Inv. USA). The threshold of significance was set to 5% ($p<0.05$).

Western blot for transglutaminase-1 (TG1) and hornerin dosages

Cell culture

Normal human keratinocytes, from a 24 years old woman, were cultured in a defined medium (Epilife CF) for 7 days in the presence or absence of different final concentrations of *Nephelium lappaceum* peel extract at a dose equivalent to Nephydrat from 0.015% up to 0.05%.

Protein analysis

After treatment, the cells were harvested and lysed with a specific lysis buffer to evaluate TG1 and hornerin concentration. Protein concentration was determined by BCA assay and the samples were kept frozen at -80°C until use. All the samples were adjusted to the same protein concentration to deposit the same quantities in each capillary. Samples and reagents were prepared according to the recommended manufacturer's instructions. Target proteins were identified by a capillary electrophoresis-based protein analysis system (ProteinSimple Sally Sue™ Instrument, ProteinSimple, San Jose, California, USA) using primary antibodies and immunoprobed using a horseradish peroxidase-conjugated secondary antibody and chemiluminescent substrate. The resulting chemiluminescent signal was detected and quantified by Compass Software version 2.7.1 (ProteinSimple).

Results and statistics

The results are expressed as a percentage, as the mean +/- standard deviation (SD) compared to the untreated control standardized to 100%. Each condition was carried out with n=3-9. The statistical analysis versus the untreated control was done after normal distribution comparison of the values (Shapiro-Wilk test) following Sigmaplot® software recommendations (Systat Software Inv. USA). The threshold of significance was set to 5% ($p<0.05$).

Elisa test for involucrin dosage

Cell culture

Normal human keratinocytes, from 33, 41 and 51 years old women, were seeded in standard culture medium MCDB153 containing a low concentration of calcium (0.03 mM) and 2% of FCS in cell culture plates previously coated by a solution of type I collagen. After incubation for 3 days at 37°C and CO_2 at 5%, the culture medium was exchanged to fresh standard culture medium (without FCS) in which *Nephelium lappaceum* peel extract at dose equivalent to Nephydrat 0.2% and 0.3% was added.

Protein analysis

After incubation for 3 days at 37°C and CO_2 at 5%, the culture medium was removed, and cell viability was measured by the MTT test. Cellular involucrin was extracted and then measured by an ELISA kit, in accordance with the recommendations of the manufacturer.

Results and statistics

The results were calculated versus a range of involucrin, and then they were expressed as a percentage versus the control cell culture medium without product (untreated control). The results are presented as a mean ± SD (standard deviation) from 3 assays from 3 different donors made in triplicate. The statistical analysis versus the untreated control was done after normal distribution comparison of the values (Shapiro-Wilk test) following Sigmaplot® software recommendations (Systat Software Inv. USA). The threshold of significance was set to 5% ($p<0.05$).

EFFICACY

RE-ENERGIZING THE CELLS THROUGH ATP AND PROTECTING THE MITOCHONDRIA AGAINST DEHYDRATION STRESS

Stimulation of ATP production

in vitro

OBJECTIVE

ATP is the molecule that carries energy within cells. At the keratinocyte level, ATP is crucial to their proliferation, differentiation, and all synthesis (proteins, lipids, enzymes…) which lead to the formation of a strong and cohesive *stratum corneum*.
In this study, we evaluated the effect of Nephydrat on ATP synthesis by keratinocytes.

RESULTS & DISCUSSION

Nephydrat significantly increased ATP synthesis 7-fold and 27-fold, at 0.3% and 0.5% respectively (Figure 14).

By stimulating ATP synthesis *in vitro*, Nephydrat fuels keratinocytes with energy, to favor a strong and cohesive *stratum corneum* and consequently well-hydrated skin.

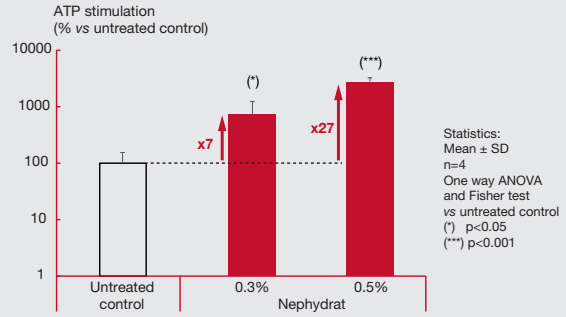


Figure 14 - Effect of Nephydrat on ATP synthesis in keratinocytes.

MATERIALS&METHODS

Cell culture

Human keratinocytes HaCaT (cell line) were seeded at 25 000 cells/cm² in a DMEM medium and incubated at 37°C 5%, CO_2, relative humidity >95% for 3 days in order to obtain a saturated cell layer. After this period, the culture medium in the microplates was replaced by a Hanks medium without (untreated control) or with different final concentrations of *Nephelium lappaceum* peel extract, at a dose equivalent to Nephydrat 0.3% and 0.5%. After 3 days of incubation and rinsing with PBS buffer, proteins were extracted with NaOH 0.1N and their content was determined with a Bradford dosage.

ATP dosage

The levels of intracytosolic ATP were measured using a bioluminescent technique (ATP Bioluminescence Assay Kit, Roche) based on the reaction of ATP with luciferin.

Results and statistics

Each value of ATP content was normalized to the corresponding protein content and finally the results are expressed as a percentage, representing the mean ± standard deviation (SD) compared to untreated control standardized to 100%. Each condition was carried out n=4. The statistical analysis versus the untreated control was done after normal distribution comparison of the values (Shapiro-Wilk test) following Sigmaplot® software recommendations (Systat Software Inv. USA). The threshold of significance was set to 5% ($p<0.05$).

Preserving mitochondrial function for energy production under dehydration stress: evaluation of the mitochondrial membrane potential

in vitro

OBJECTIVE

Mitochondrial membrane potential (MiMP) is an indicator of mitochondrial health and viability (Zorova *et al.*, 2018). Depolarized mitochondria with low miMP is observed when the cells are depleted of energy with subsequent death (Joshi and Bakowska, 2011).

The goal of this study is to evaluate the effect of Nephydrat on mitochondrial membrane potential from osmotically-stressed (dehydration-stressed) human keratinocytes.

RESULTS & DISCUSSION

As shown on Figure 15, mannitol at 300 mM induced osmotic dehydration stress. It significantly increased the proportion of keratinocytes with dysfunctional mitochondria (depolarized mitochondria with low MiMP). Nephydrat at 0.1% preserved keratinocyte mitochondria from the harmful effect of osmotic dehydration stress, by significantly reducing the proportion of dysfunctional mitochondria by 69%. The level of dysfunctional mitochondria thus reached was similar to the initial untreated control condition level.

With Nephydrat, keratinocytes' mitochondria are preserved while exposed to an osmotic dehydrating shock *in vitro*.

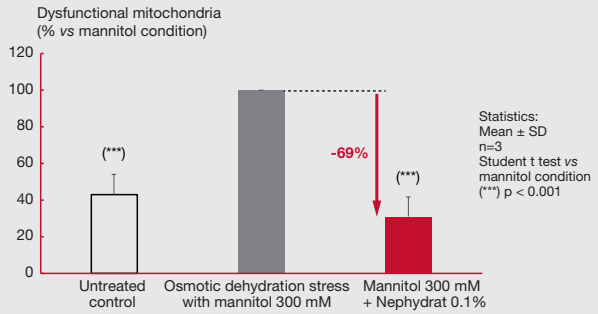


Figure 15 - Effect of Nephydrat at 0.1% on the mitochondrial membrane potential of mitochondria (MiMP) from keratinocytes stressed with an osmotic dehydration shock (mannitol 300 mM). A low mitochondrial membrane potential assay corresponds to cells with "dysfunctional mitochondria".

MATERIALS&METHODS

Cell culture

Human keratinocytes HaCaT (cell line) were seeded at 25 000 cells/cm² within a standard medium containing FCS and incubated at 37°C, 5% CO_2, relative humidity >95% for 3 days to obtain a saturated cell layer. The cell culture medium was changed to a standard medium (DMEM) without (untreated control) or with *Nephelium lappaceum* peel extract, at a dose equivalent to Nephydrat 0.1% and cells were incubated for 1 day at 37°C. Then, the cell culture medium was changed to a standard medium (DMEM) containing 300 mM mannitol and 300 mM mannitol with *Nephelium lappaceum* peel extract, at a dose equivalent to Nephydrat 0.1%. The untreated control corresponded to the cells without any added products. Cells were then incubated for 1 day at 37°C.

Mitochondrial membrane potential (MiMP)

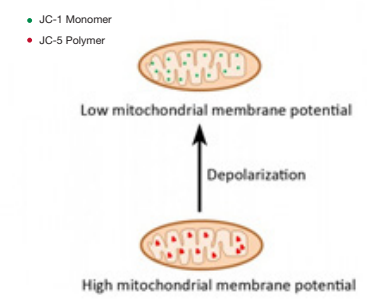


Figure 16 – Mitochondrial membrane potential measurement with JC-1 probe.

Mitochondrial activity was measured through a fluorescent tracer JC-1 (Sigma). The JC-1 probe is a lipophilic cationic dye able to selectively enter mitochondria. JC-1 penetration in mitochondria is directly dependent of MiMP. So, when MiMP is high, JC-1 molecules are accumulated in mitochondria where they form an aggregate characterized by a specific red fluorescence. Conversely when MiMP is low, JC-1 molecules are diluted in the cell cytoplasm where isolated molecules produce a specific green fluorescence (Cottet-Rousselle et al., 2011). Hence, healthy cells with hyperpolarized mitochondrial fluorescence were mainly in red while damaged (apoptotic) cells with dissipated MiMP fluorescence were mainly in green (Figure 16).

Suspensions of keratinocytes were prepared by trypsinization and stained by JC-1 probe at 1 µM in Hank's solution. Then, the cell suspensions were rinsed by Hanks solution and analyzed by flow cytometry equipped with a blue laser (excitation at 488 nm). The green fluorescence (at 530 nm) was recorded in channel 1 while red fluorescence (at 595 nm) was recorded in channel 2.

Results and statistics

Results are expressed as the percentage of green fluorescent cells (= dysfunctional mitochondria) referring to total cell population and presented as the mean ± standard deviation from 3 assays. The statistical analysis versus the mannitol condition was done after normal distribution comparison of the values (Shapiro-Wilk test) following Sigmaplot® software recommendations (Systat Software Inv. USA). The threshold of significance was set to 5% ($p<0.05$).

Preserving the mitochondrial network structure for energy production

in vitro

OBJECTIVE

Mitochondria can change their shape from discrete isolated organelles to a large continuous reticulum. The cellular advantages underlying these fused networks are still incompletely understood (Hointzing *et al.*, 2015). However, mitochondrial fragmentation occurs in response to cellular dysfunction and apoptosis (Arnoult, 2007; Karbowski and Youle, 2003).

The goal of this study is to evaluate the effect of Nephydrat on the mitochondrial network in keratinocytes submitted to dehydration stress.

RESULTS & DISCUSSION

Figure 17A shows the mitochondrial network in keratinocytes in the untreated condition: long, continuous and fused filaments were observed. In the presence of mannitol, a disturbed mitochondrial network was observed with fragmented filaments (Figure 17B). With Nephydrat at 0.1%, an improvement of the mitochondrial network was observed, with a decrease of network fragmentation and recovery of continuous filaments (Figure 17C).

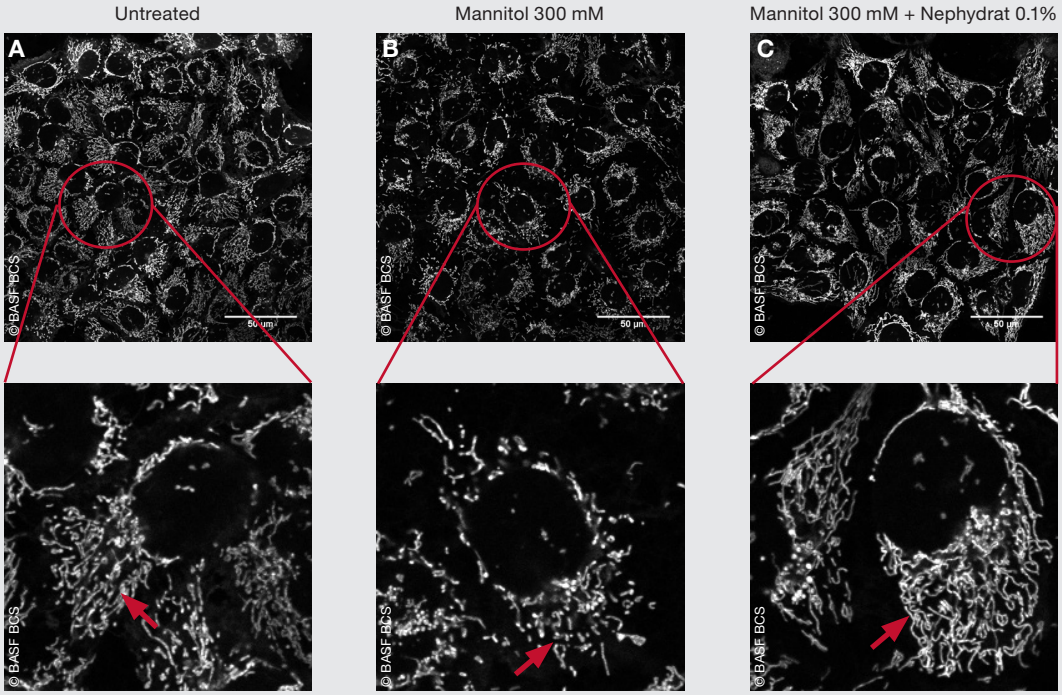


Figure 17 - Microscopic observations of the mitochondrial network structure in keratinocytes. Mitochondrial networks are visualized in grey levels. Dark circular shapes correspond to the keratinocyte nuclei. **A** - Untreated control, with long, continuous and fused filaments. **B** - Cells stressed with 300 mM mannitol: fragmented filaments are observed. **C** - Cells stressed with 300 mM mannitol and treated with Nephydrat 0.1%: long and continuous filaments are preserved.

With Nephydrat, keratinocytes' mitochondrial network is preserved while exposed to osmotic dehydration stress *in vitro*.

CONCLUSION

With Nephydrat, highly energized mitochondria can maintain a network of fused filaments, in order to enable better control of the production and distribution of energy in cells.

MATERIALS&METHODS

Cell culture

Human keratinocytes HaCaT (cell line) were seeded at 8000 cells/cm² within a Hanks medium and incubated at 37°C, CO_2 5%, relative humidity >95% for 3 days to obtain a saturated cell layer. The cell culture medium was exchanged to a balanced salt solution of Hanks containing *Nephelium lappaceum* peel extract - at a dose equivalent to Nephydrat 0.1% - and 300 mM mannitol. The untreated control corresponded to the cells without any added products. Cells were then incubated for 4 hours at 37°C. Then the balanced salt solutions are exchanged for a solution of MitoTracker™ Green FM (Life Technologies M7514) to reveal active mitochondria. After rinsing with a balanced solution of Hanks, the cell layers are observed with fluorescence microscopy.

Mitochondrial network imaging

A confocal microscope (LSM780, Zeiss) with an incubation box was used. Cells were excited with a 488 nm laser and the emitted signal from 495 nm to 630 nm was collected. Cells were observed with a 40X/1.4 Oil DIC M27 objective (Zeiss). One plane was imaged with the best focus on the mitochondrial network and 6 areas per condition were imaged to have a good distribution over the network.

EFFICACY

STIMULATING THE INTRA-CELLULAR UPTAKE OF TAURINE FOR CELL PROTECTION AGAINST DEHYDRATION

in vitro

OBJECTIVE

Taurine is an important osmolyte required for keratinocyte hydration. Under stress conditions, taurine allows cells to maintain constant volume and intracellular ionic strength (Anderheggen *et al.*, 2006). The taurine transporter (TauT) is necessary for taurine to enter the cells. Once inside keratinocytes, taurine allows the re-diffusion of water into the cells to maintain a constant volume and their correct functioning.

The aim of this study was to evaluate, on human keratinocytes, the effect of Nephydrat on TauT (Taurine transporter) and taurine intracellular content.

RESULTS & DISCUSSION

As shown on Figure 18A, Nephydrat at 0.5% significantly increased TauT level by 44%. Consequently, taurine intracellular content significantly increased by 50% and 81% with Nephydrat treatment at 0.1% and 0.5% respectively (Figure 18B).

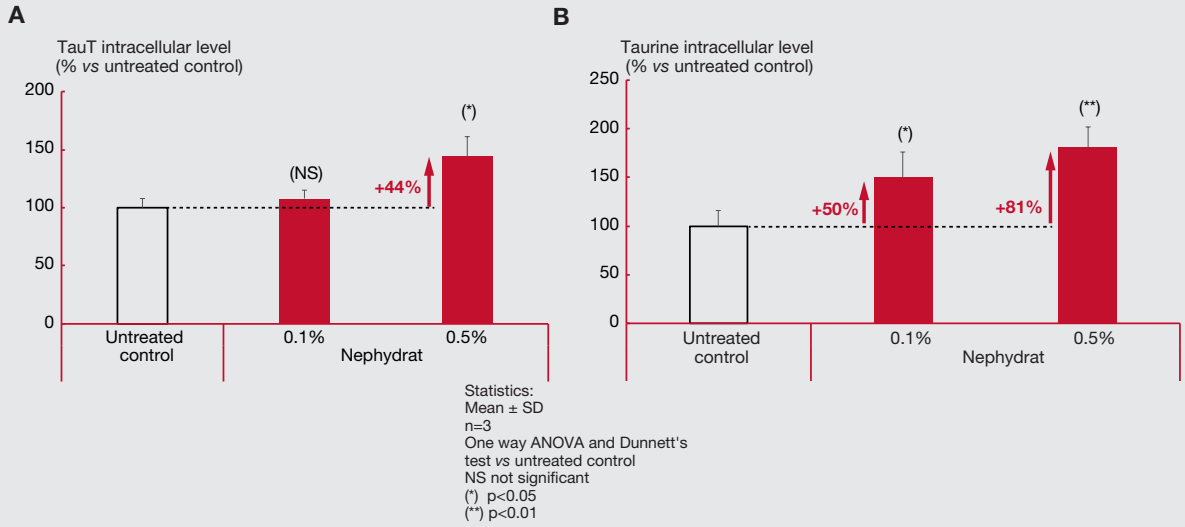


Figure 18 - Effect of Nephydrat on TauT synthesis (A) and taurine intracellular content (B)

CONCLUSION

Nephydrat stimulates TauT synthesis, which facilitates taurine uptake by keratinocytes. Then, taurine attracts water into the cells allowing keratinocytes to preserve their volumes, which is crucial for the maintenance of optimal cell functions.

MATERIALS&METHODS

Cell culture

Human keratinocytes were obtained from a 50 years old donor. Keratinocytes were cultivated in a monolayer until reaching confluency. Cells were incubated for 24 hours in the absence (control) or in the presence of different final concentrations of *Nephelium lappaceum* peel extract, at dose equivalent to Nephydrat 0.1 and 0.5%.

TauT and Taurine dosage

At the end of the incubation period, intracellular taurine transporter and intracellular taurine content were quantified in cell lysate (obtained by sonication) using a specific ELISA assay.

Results and statistics

The results are expressed as a percentage, representing the mean ± standard deviation (SD) compared to the untreated control standardized to 100%. Each condition was carried out n=3. The level of significance between "untreated control" and "Nephydrat" was assessed independently by a one factor variance analysis (One-way ANOVA) followed by a Holm-Sidak test. The threshold of significance was set to 5% ($p<0.05$).



Conclusions *in vitro*

We demonstrated the contributing effects of Nephydrat on:

Strengthening the *stratum corneum*, by acting on:
- the cement: stimulation of phospholipids and ceramides including long chain ceramides,
- the bricks: stimulation of involucrin, hornerin, transglutaminase-1 and long chain ceramides.

Optimization of cell energy:
- re-energizing the cells through ATP,
- protection of mitochondria when exposed to dehydration stress.

Stimulation of the intra-cellular uptake of taurine for protection against dehydration stress.

Nephydrat acts on several targets linked to hydration, for higher efficacy.

EFFICACY

Two clinical studies were run to evaluate Nephydrat efficacy: an instrumented study, measuring skin hydration with the MoistureMap method and a panel assessment study done on 3 continents, with dedicated finished product formulations.

IMPROVEMENT OF SKIN HYDRATION

In vivo

OBJECTIVE

In a double-blind, randomized, hemi-face, placebo-controlled clinical study, we evaluated the ability of Nephydrat to improve skin hydration (Figure 19).

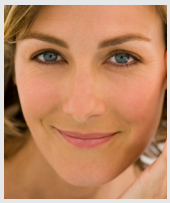


Figure 19 - Clinical study on skin hydration with Nephydrat *vs.* placebo: skin hydration evaluated with MoistureMap method.

The study was performed on 27 Caucasian female volunteers aged from 29 to 46, with dry skin on the face.
Nephydrat at 2% and a placebo formulation were applied twice a day for a period of 28 days. The measurements were done at D0, D14 and D28.

RESULTS & DISCUSSION

From 28 days of treatment with a formulation containing Nephydrat at 2%, the hydration improvement increased significantly by 7% *vs.* baseline ($p<0.001$) while the placebo formulation showed no significant difference (Figure 20). This change is higher than with the placebo, even though the difference between the two products' application is not statistically significant.

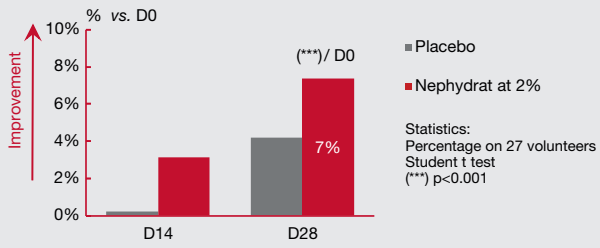


Figure 20 - Percentage of skin hydration improvement *vs.* baseline measured with MoistureMap equipment.

Pictures in Figure 21 illustrate the improvement of skin hydration with 2% Nephydrat. Darker pixels represent more hydrated areas. Visible improvement is observed after 28 days of treatment with 2% Nephydrat compared with the placebo.

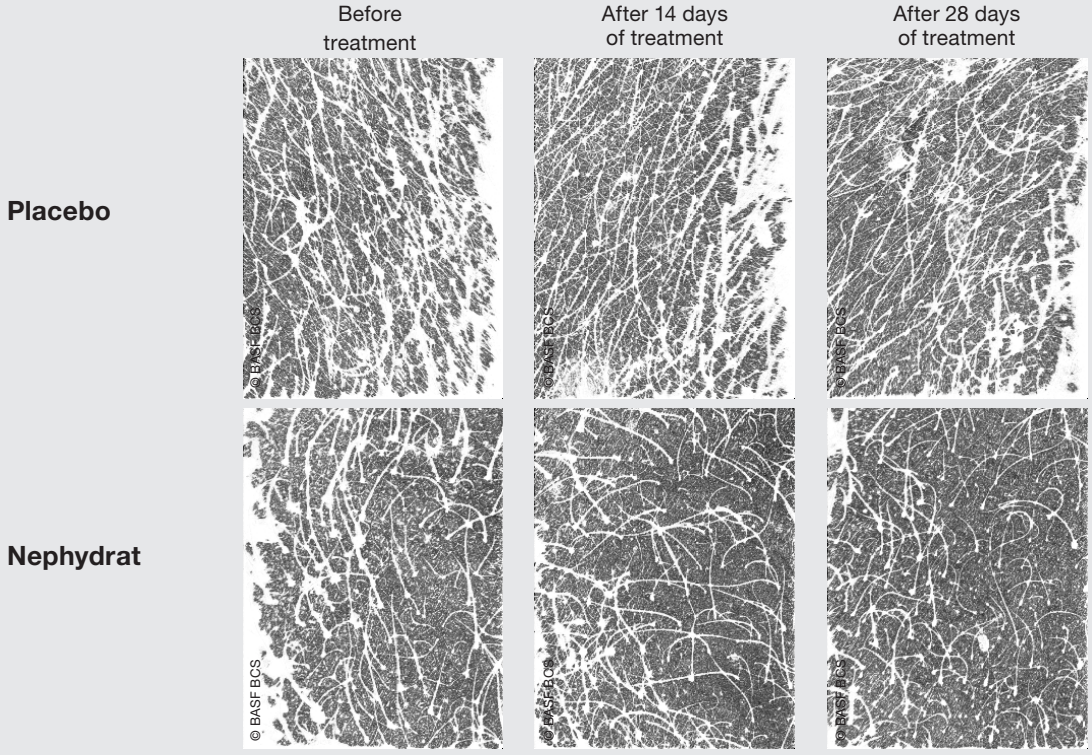


Figure 21 - Illustrative images of one volunteer with MoistureMap equipment.

The increase of cutaneous hydration as compared to baseline demonstrated that, after 28 days, the application of a formulation with Nephoria at 2% promoted an improvement in skin moisturization.

PANEL ASSESSMENT OF FORMULATIONS

in vivo

OBJECTIVE

Using self-assessment questionnaires, we evaluated the volunteers' perception of formulations containing 2% Nephydrat to improve skin hydration. These panel assessments were conducted in Europe, Asia and North America using 3 different adapted formulations.

The first study was done on 31 Caucasian female volunteers in Lyon (France), the second on 30 Caucasian female volunteers in Montreal (Canada) and the third on 33 Asian female volunteers in Bangkok (Thailand).

For each region, a specific formulation containing Nephydrat at 2% was designed: a cocooning emulsion for the European panel (annex 3), a comfort gel cream for the North American panel (annex 4) and a smoothing light emulsion for the Asian panel (annex 5). Each formulation was applied twice a day for a period of 14 days. Then a self-assessment questionnaire focused on organoleptic and efficacy evaluation was filed by the volunteers.

RESULTS & DISCUSSION

A significant majority of the volunteers approved the formulation specifically developed for their region (Figure 22 A&B). The texture and the sensorial feeling once applied was particularly appreciated (non-greasy and non-sticky).

A

	Europe (Lyon, France, Caucasians)		North America (Montreal, Canada, Caucasians)		Asia (Bangkok, Thailand, Asians)	
	%	Statistics p value	%	Statistics p value	%	Statistics p value
The texture of the product is pleasant	100%	<0.05	97%	<0.05	100%	<0.05
The fragance of the product is pleasant	81%	<0.05	80%	<0.05	91%	<0.05
Speed of penetration	97%	<0.05	90%	<0.05	94%	<0.05
Non-greasy	100%	<0.05	93%	<0.05	97%	<0.05
Non-sticky	94%	<0.05	97%	<0.05	97%	<0.05

B

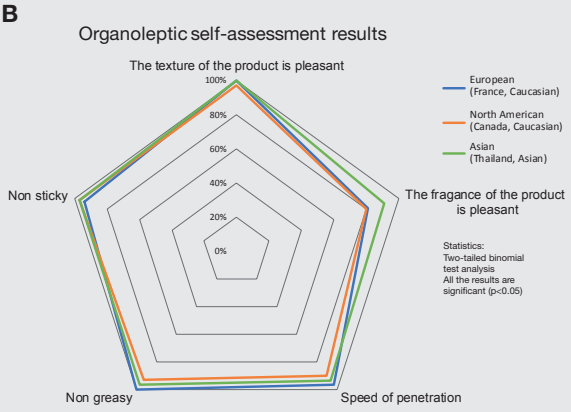


Figure 22 - Organoleptic self-assessment results: percentage of favorable opinion for formulations containing Nephydrat at 2% after 14 days. Statistics: two-tailed binomial test. **A** - Result table. **B** - Radar graph presentation.

Concerning the efficacy, after 14 days of application, a significant majority of volunteers perceived an increase in skin hydration, an improvement of skin softness and radiance with each of the 3 different 2% Nephydrat formulations (Figure 23 A&B). In addition, the North American and Asian volunteers perceived, with a significant majority, a skin boosting effect (my skin is more luminous & the formulation gives a "boost effect" to my skin).

A

	Europe (Lyon, France, Caucasians)		North America (Montreal, Canada, Caucasians)		Asia (Bangkok, Thailand, Asians)	
	%	Statistics p value	%	Statistics p value	%	Statistics p value
My skin is more luminous	68%	NS	80%	<0.05	88%	<0.05
The formulation gives a "boost effect" to my skin	65%	NS	83%	<0.05	100%	<0.05
My skin is more hydrated	94%	<0.05	83%	<0.05	100%	<0.05
My skin is protected from dehydration	81%	<0.05	80%	<0.05	97%	<0.05
My skin is softer	81%	<0.05	90%	<0.05	100%	<0.05
The formulation restores radiance to the skin	71%	<0.05	83%	<0.05	88%	<0.05

B

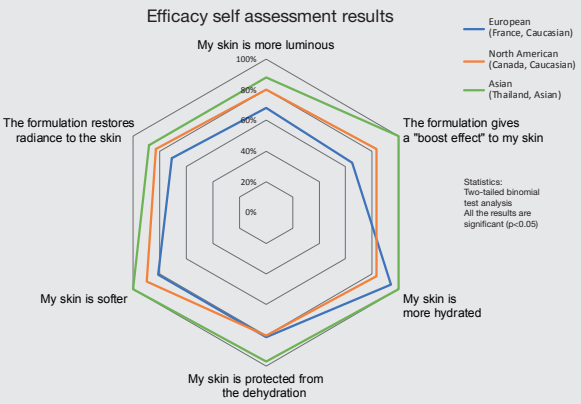


Figure 23 - Efficacy self-assessment results: percentage of favorable opinion for formulations containing Nephydrat at 2% after 14 days. Statistics: two-tailed binomial test. **A** - Result table. **B** - Radar graph presentation.

CONCLUSION

Nephydrat helps to increase skin hydration.

The benefits to skin hydration, softness and radiance of formulations containing Nephydrat are perceivable by European, Asian and North American consumers.

MATERIALS&METHODS

MoistureMap study

Study design

The clinical study was carried-out randomized and double-blinded versus placebo during winter (in February). The efficacy of the formulation containing Nephydrat at 2% was compared to the baseline (before treatment, D0) and to the half-face treated with the placebo formulation. The formulation is detailed in Annex 2. The study was conducted for a period of 28 days with check points at D0, D14 and D28.

Inclusion criteria

The study was done on 27 Caucasian female volunteers, between 29 and 46 years old and with dry skin on the face (cutaneous hydration rate < 100 A.U. on jaws, checked using MoistureMap).

Application modality

The products (a formulation containing Nephydrat at 2% or a placebo formulation) were applied on the face by the volunteers twice a day on each half face for 28 days, under normal conditions of use.

Evaluation methods

The MoistureMap MM100 (Courage & Khazaka) provides a non-invasive, fast and precise measurement, allowing a visualization of the surface hydration state of the studied area.

This equipment measures the penetration of the electromagnetic field by the technique of “Capacitance Imaging”. This technology has been used for several years to record fingerprints in the security services. The images from the sensor give graphical information on the near surface hydration distribution and the microtopography of the tissue. The probe has a size of 4.3 x 3 cm and a measurement surface of 18 x 12.8 mm with more than 90 000 micro-capacitors located every 50 µm. This configuration allows a very precise visualization of the hydration topography on the analyzed area. Capacitance values are transformed into pixels, on a scale of 256 gray levels with darker pixels representing the more hydrated areas.

The measurement was made on the cheek just above the jawbone. An increase in the “cutaneous hydration index” parameter shows the product’s moisturizing effect.

Statistics

Results are expressed as the mean percentage of change compared to the baseline measurement.

The statistical analysis of the evolution of the parameters as a function of time and of the differences in the studied parameters between the treatment groups was done after verifying the normality of distribution using the Shapiro-Wilk test. Afterwards, the statistical analysis of the evolution of the studied parameters for each product was performed with the Student t test after the normality of distribution had been confirmed.

Panel Self-assessment study

Study design

The panel assessments were conducted in Europe, North America and Asia. Specific formulations were designed for each region (see Annex 3, 4 and 5). The self-assessment questionnaires were filed after 14 days of twice daily application of each formulation containing Nephydrat at 2%.

Inclusion criteria

The European panel assessment was done on 31 Caucasian female volunteers in France (Lyon), the North American assessment on 30 Caucasian female volunteers in Canada (Montreal) and the Asian assessment on 33 Asian female volunteers in Thailand (Bangkok). All volunteers presented dry skin on the face (cutaneous hydration rate < 100 A.U. on jaws, checked using MoistureMap).

Application modality

The formulations containing Nephydrat at 2% were applied twice a day for 14 days, under normal conditions of use

Self-assessment questionnaire

The questionnaire analysis involved establishing frequency tables that take into account the number of responses and calculate the frequency of the different possible answers (given as percentage) to each qualitative question. To evaluate the efficacy and the appreciation of the products for each item, two percentages Z1 and Z2 were calculated as follows:

- Z1 = favorable opinion (“completely agree” + “somewhat agree”),
- Z2 = unfavorable opinion (“completely disagree” + “somewhat disagree”).

The statistical difference in frequencies (%) between favorable and unfavorable opinions was evaluated using two-tailed binomial test at 5%.



GENERAL CONCLUSION

Moisturization is a top consumer need and the top claim in the skin care market. However, our lifestyles have evolved, and we must not forget that our day-to-day choices may potentially impact the future of our planet. Taking up the trends of modern consumer demand and their growing environmental awareness, has prompted the cosmetic industry to develop more sustainable solutions, while keeping the same high level of demonstrated *in vitro* biological efficacy and clinical performance.

Building on the experience acquired with our previous sustainable and responsible argan program in Morocco, we have created a new socially responsible supply chain in cooperation with a long-standing local partner in Vietnam. We identified rambutan (*Nephelium lappaceum*) as the most promising resource.

Exploring the different parts of the rambutan tree, we discovered that Nephydrat, a specifically selected extract of the protective envelope of the rambutan fruits (pericarp), is able to rescue dry and weakened skin with a 3-step strategy.

Firstly, we demonstrated *in vitro* that Nephydrat reinforced both the cement (lipophilic extracellular medium) and brick (corneocyte) components of the *stratum corneum*. Nephydrat helped to increase the synthesis of phospholipids and ceramides, including long chain ceramides. In addition, Nephydrat stimulated both involucrin, hornerin, transglutaminase -1 for a strengthened and cohesive *stratum corneum*. Secondly, Nephydrat fueled keratinocytes with energy (ATP) *in vitro*, and protected mitochondria, the cell “powerhouses”, against dehydration stress. Finally, Nephydrat stimulated the cellular uptake of taurine, which is crucial to maintaining optimal cellular hydration. All these *in vitro* results show that Nephydrat acts on hydration through multiple targets.

Tested *in vivo*, Nephydrat at 2% improved skin hydration within 1 month. In independent self-assessment studies conducted in Europe, Asia and North-America, consumers in each region using 2% Nephydrat formulations confirmed the benefits for their skin after 2 weeks, with improvement to skin hydration, softness and radiance.

While the juicy rambutan fruit is kept from dehydration by its spiny-peel, our skin is protected from dehydration thanks to its protective envelope, the epidermis. The protective power of the rambutan peel is extracted to create Nephydrat, a hydro-energizer active ingredient that preserves the skin from dehydration.

ANNEXES

Annex 1 - Technical data - Available upon request

- Quality and Regulatory Product Information
- Composition sheet
- Specifications
- Formulation Data Sheet

Annex 2 - Clinical test formula

Trade name	INCI name	Placebo formulation %	Nephydrat formulation %
Emulgade PL	Cetearyl Glucoside (and) Cetearyl Alcohol	2.00	2.00
Cutina® HVG	Hydrogenated Vegetable Glycerides	2.00	2.00
Cetiol® Sensoft	Propylheptyl Caprylate	5.00	5.00
Cegesoft® PS 6	Olus Oil	3.00	3.00
Cetiol® 4 All	Dipropylheptyl Carbonate	4.00	4.00
Xiameter PMX-200 Silicone Fluid 50 CS (Dow Corning)	Dimethicone	1.00	1.00
Cosmedia® SP	Sodium Polyacrylate	0.70	0.70
Elestab™ 388	Propylene Glycol (and) Phenoxyethanol (and) Chlorphenesin (and) Methylparaben	2.50	2.50
Glycerin 99-5	Glycerin	2.60	1.00[a]
Eumulgin® SG	Sodium Stearoyl Glutamate	0.40	0.40
Cetiol® Ultimate	Undecane (and) Tridecane	4.00	4.00
Citric acid solution 10%	Citric acid (and) Water	0.60	0.60
Nephydrat BC 10061	Glycerin (and) Water (and) Nephelium Lappaceum Peel Extract	-	2.00
Water, demin	Aqua	qsf 100	qsf 100

[a] The content in glycerin was adjusted to have exactly the same content in the two formulations: 2.6%

Annex 3 - Cocooning emulsion (SC-FR-19-BC-50829-02)

Phase	Ingredients	INCI	% by weight	Function
A	Emulgade® PL 68/50	Cetearyl Glucoside, Cetearyl Alcohol	2.00	Emulsifier (O/W)
	Cutina® HVG	Hydrogenated Vegetable Glycerides	2.00	Consistency agent
	Cetiol® Sensoft	Propylheptyl Caprylate	5.00	Emollient
	Cegesoft® PS 6	Olus Oil [EU], Vegetable Oil [CTFA]	3.00	Emollient
	Cetiol® 4 All	Dipropylheptyl Carbonate	4.00	Emollient
	Xiameter PMX-200 Silicone Fluid 50CS *(Dow Corning)*	Dimethicone	1.00	Defoamer
	Cosmedia® SP	Sodium Polyacrylate	0.70	Rheology modifier
B	Water, demin.	Aqua	74.30	
	Preservative		q.s.	Preservative
	Glycerin	Glycerin	1.00	Humectant
	Eumulgin® SG	Sodium Stearoyl Glutamate	0.40	Emulsifier (O/W)
C	Cetiol® Ultimate	Undecane, Tridecane	4.00	Emollient
D	Nephydrat™ BC10061	Glycerin, Nephelium Lappaceum Peel Extract	2.00	Active ingredient
E	Citric Acid (10% solution)	Citric Acid	0.60	pH Adjustment

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Annex 4 - Comfort gel cream (SC-FR-19-BC-50859-02)

Phase	Ingredients	INCI	% by weight	Function
A	Water, demin.	Aqua	74.52	
	Elestab® 388	Propylene Glycol, Phenoxyethanol, Chlorphenesin, Methylparaben	2.50	Preservative
	Cosmedia® Ace	Sodium Polyacrylate, Dicaprylyl Carbonate, Polyglyceryl-3 Caprate	1.00	Rheology modifier
B	Myritol® 318	Caprylic/Capric Triglyceride	6.66	Emollient
	Cetiol® C 5C	Coco-Caprylate/Caprate	6.66	Emollient
	Cegesoft® PS 6	Olus Oil [EU], Vegetable Oil [CTFA]	6.66	Emollient
C	Nephydrat™ BC10061	Glycerin, Nephelium Lappaceum Peel Extract	2.00	Active ingredient

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Annex 5 - Smoothing light emulsion (SC-FR-19-BC-50802-08)

Phase	Ingredients	INCI	% by weight	Function
A	Emulgade® Sucro Plus	Sucrose Polystearate, Cetyl Palmitate	3.00	Emulsifier (O/W)
	Cutina® PES	Pentaerythrityl Distearate	1.00	Consistency agent
	Myritol® 318	Caprylic/Capric Triglyceride	3.00	Emollient
	Cetiol® C 5C	Coco-Caprylate/Caprate	3.00	Emollient
	Cetiol® CC	Dicaprylyl Carbonate	3.00	Emollient
	Cosmedia® SP	Sodium Polyacrylate	0.70	Rheology modifier
B	Water, demin.	Aqua	82.80	
	Glycerin	Glycerin	0.40	Humectant
	Eumulgin® SG	Sodium Stearoyl Glutamate	0.50	Emulsifier (O/W)
	Preservative		q.s.	Preservative
C	Nephydrat™ BC10061	Glycerin, Nephelium Lappaceum Peel Extract	2.00	Active ingredient
D	Citric Acid (10% solution)	Citric Acid	0.60	pH Adjustment

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Annex 6 - Cung Cap Nuoc Gel Cream (SC-DE-18-151-15)

Phase	Ingredients	INCI	% by weight	Function
A	Water, demin.	Aqua	56.80	
	Glycerin	Glycerin	8.00	Humectant
	1,3-Butanediol	Butylene Glycol	3.00	Humectant
	Euxyl K 712 *(Schülke)*	Aqua, Sodium Benzoate, Potassium Sorbate	1.00	Preservative
	Puricolor® Red FRE1 FDA (0.1% Solution)	CI 14700	0.05	Colorant
	Ethanol	Alcohol	8.00	Cooling agent
	Nephydrat™ BC10061	Glycerin, Nephelium Lappaceum Peel Extract	2.00	Active ingredient
B	Rheocare® XGN	Xanthan Gum	0.10	Stabilizer
	Cosmedia® Ace	Sodium Polyacrylate, Dicaprylyl Carbonate, Polyglyceryl-3 Caprate	3.00	Rheology modifier
C	Eumulgin® SG	Sodium Stearoyl Glutamate	0.05	Emulsifier (O/W)
	Cetiol® SB 45	Butyrospermum Parkii Butter	2.00	Emollient
	Cegesoft® VP	Olus oil, Hydrogenated Vegetable Oil, Candelilla cera [EU], Vegetable Oil, Hydrogenated Vegetable Oil, Euphorbia Cerifera (Candelilla) Wax	2.00	Emollient
	Myritol® 318	Caprylic/Capric Triglyceride	8.00	Emollient
	Cetiol® Ultimate	Undecane, Tridecane	5.00	Emollient
	Copherol® F 1300 C	Tocopherol	0.30	Antioxidant
	Perfume*	Parfum	0.20	Fragrance
D	Citric Acid (20% solution)	Citric Acid	0.50	pH Adjustment

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BIBLIOGRAPHY

Anderheggen B, Jassoy C, Waldmann-laue M, Förster T, Wadle A, Doering T.
Taurine improves epidermal barrier properties stressed by surfactants. A role for osmolytes in barrier homeostasis.
J. Cosmet. Sci., 57:1-10, 2006

Arnoult D.
Mitochondrial fragmentation in apoptosis.
Trends cell biol., 17:1,6-12, 2007

Bouwstra JA, Gooris GS, Dubbelaar FE, Ponec M.
Cholesterol sulfate and calcium affect *stratum corneum* lipid organization over a wide temperature range.
J. Lipid Res., 40(12):2303-12, 1999

Candi E, Schmidt R, Melino G.
The cornified envelope: a model of cell death in the skin.
Nat. Rev. Mol. Cell. Biol, 6(4):328-40, 2005

Codech L, Lopez O, De la Maza A, Parra JL.
Ceramides and skin function.
Am. J. Clin. Dermatol., 4:2, 2003

Cottet-Rousselle C, Ronot X, Leverve X, Mayol JF.
Cytometric assessment of mitochondria using fluorescent probes.
Cytometry Part A, 79A:405-425, 2011

Elias PM, Gruber R, Crumrine D, Menon G, Williams ML, Wakefield JS, Holleran WM, Uchida Y.
Formation and functions of the corneocyte lipid envelope (CLE).
Biochim. Biophys. Acta., 1841:314-18, 2014

Elias PM, Williams ML, Choi EH, Feingold KR.
Role of cholesterol sulfate in epidermal structure and function: Lessons from X-linked ichthyosis.
Biochim. Biophys. Acta., 1841(3):353-61, 2014

Feichtinger RG, Sperl W, Bauer JW, Kofler B.
Mitochondrial dysfunction: a neglected component of skin diseases.
Exp. Dermatol., 23:607-614, 2014

Feingold K, Elias PM.
Role of lipids in the formation and maintenance of the cutaneous permeability barrier.
Biochim. Biophys. Acta,1841:280-294, 2014

Fluhr JW, Kao J, Jain M, Kenneth SKA, Feingold K, Elias PM.
Generation of free fatty acids from phospholipids regulates *stratum corneum* acidification and integrity.
J. Invest. Dermatol., 117:44-51, 2001

Guneri D, Voegeli R, Gurgul SJ, Munday MR, Lane ME, Rawlings AV.
A new approach to assess the effect of photodamage on corneocyte envelope maturity using combined hydrophobicity and mechanical fragility assays.
Int. J. Cosmetic. Sci., 40:207-216, 2018

Grafe F, Wohlrab W, Neubert RH, Brandsch M.
Functional characterization of sodium- and chloride-dependent taurine transport in human keratinocytes.
Eur. J. Pharm. Biopharm., 57:337-341, 2004

Henry J, Hsu CY, Haftek M, Nachat R, de Koning HD, Gardinal-Galera I, Hitomi K, Balica S, Jean-decoster C, Schmitt AM, Paul C, Serre G, Simon M.
Hornerin is a component of the epidermal cornified cell envelopes.
FASEB J., 25;5,1567-1576, 2011

Hernandez-Hernandez C, Aguilar CN, Rodriguez-Herrera R, Flores-Gallegos AC, Morlett-Chavez J, Govea-Salas M, Ascacio-Valdes JA.
Rambutan (*Nephelium Lappaceum* L.): nutritional and functional properties.
Trends in food science & technology, 85, 201-210, 2019

Hoitzing H, Johnston LG, Jones NS.
What is the function of mitochondrial networks? A theoretical assessment of hypotheses and proposal for future research.
Bioessays, 37:687-700, 2015

Hsu C, Gasc G, Raymond AA, Burlet-Schiltz O, Takahara H, Serre G, Mechin MC, Simon M.
Deimination of human hornerin enhances its processing by calpain-1 and its cross-linking by transglutaminases.
J. Invest. Dermatol., 137:422-429, 2017

Janeke G, Siefken W, Carstensen S, Springmann G, Bleck O, Steinhart H, Höger P, Wittern KP, Wenck H, Stäb F, Sauermann G, Schreiner V, Doering T.
Role of taurine accumulation in keratinocyte hydration.
J. Invest. Dermatol., 121:354-361, 2003

Joshi DC, Bakowska JC.
Determination of Mitochondrial Membrane Potential and Reactive Oxygen Species in Live Rat Cortical Neurons.
J. Vis. Exp., 23:51, 2011

Karbowski M, Youle RJ.
Dynamics of mitochondrial morphology in healthy cells and during apoptosis.
Cell Death Differ., 10:870-880, 2003

Kudryavtseva AV, Krasnov GS, Dmitriev AA, Alekseev B., Kardymon OL, Sadritdinova AF, Fedorova MS, Pokrovsky AV, Melnikova NV, Kaprin AD, Moskalev AA, Snezhkina AV.
Mitochondrial dysfunction and oxidative stress in aging and cancer.
Oncotarget, 7:29, 2016

Lourenço R, Camilo ME.
Taurine: a conditionally essential amino acid in humans? An overview in health and disease.
Nutr. Hosp., 17:6,262-270, 2002

Mao-Qiang M, Elias PM, Feingold K.
Fatty acids are required for epidermal permeability barrier function.
J. Clin. Invest., 92:7191-798, 1993

Marekov L, Steinert PM.
Ceramides are bound to structural proteins of the human foreskin epidermal cornified cell envelope.
J. Biol. Chem., 273:28, 17763-17770, 1998

Moore DJ, Rawlings AV.
The Chemistry, function and (patho)physiology of *stratum corneum* barrier ceramides.
Int. J. Cosmetic. Sci., 39: 366-372, 2017

Nemes Z, Marekov LN, Fesus L, Steinert PM.
A novel function for transglutaminase 1: Attachment of long chain hydroxyceramides to involucrin by ester bond formation.
Proc. Natl. Acad. Sci., 96:8402-8407, 1999

Nemes Z, Steinert PM.
Bricks and mortar of the epidermal barrier.
Exp. Mol. Med., 31:1, 5-19, 1999

Pappas A.
Epidermal surface lipids.
Dermatoendocrinol., 1:2, 72-76, 2009

Proksch E, Brandner JM, Jensen J.
The skin: an indispensable barrier.
Exp. Dermat., 17:1063-1072, 2008

Rabionet M, Gorgas K, Sandhoff R.
Ceramide synthesis in the epidermis.
Biochim. Biophys. Acta, 1841:422-434, 2014

Ripps H, Shen W.
Review: Taurine: A “very essential” amino acid.
Mol. Vis., 18:2673-2686, 2012

Skulachev VP.
Mitochondrial filaments and clusters as intracellular power transmitting cables.
Trends Biochem. Sci., 26:1, 2001

Slominski AT, Brozyna AA, Tuckey TC.
Cutaneous glucocorticoidogenesis and cortisol signaling are defective in psoriasis.
J. Invest. Dermatol., 137:1609-1611, 2017

Sochorova M, Audrlicka P, Cervena M, Kovacik A, Kopecna M, Opalka L, Pullmannova P, Vavrova K.
Permeability and microstructure of cholesterol-depleted skin lipid membranes and human *stratum corneum.*
J. Colloid Interface Sci., 1;535:227-238, 2019

Uchida Y.
Ceramide signaling in mammalian epidermis.
Biochim. Biophys. Acta., 1841(3):453-462, 2014

Van Smeden J, Bouwstra JA.
Stratum corneum lipids: their role for the skin barrier function in healthy subjects and atopic dermatitis patients.
Curr. Probl. Dermatol., 49:8-26, 2016

Wertz PW.
Lipids and barrier function of the skin.
Acta Derm. Venereol., 80:208, 2000

Worden JA, Stipanuk MH.
A comparison by species, age and sex of cysteinesulfinate decarboxylase activity and taurine concentration in liver and brain of animals.
Comp. biochem. physiol. B., 82:2,233-239, 1985

Zamzami N, Marchetti P, Castedo M, Decaudin D, Macho A, Hirsch T, Susin SA, Petit PX, Mignotte B, Kroemer G.
Sequential reduction of mitochondrial transmembrane potential and generation of reactive oxygen species in early programmed cell death.
J. E. M., 182, 2:1, 367-377, 1995

Zorova LD, Popkov VA, Plotnikov EY, Silachev DN, Pevzner IB, Jankauskas SS, Babenko VA, Zorov SD, Balakireva AV, Juhaszova M, Sollott SJ, Zorov DB.
Mitochondrial membrane potential.
Anal. Biochem., 552:50-59, 2018

Europe

BASF Beauty Creations

49, avenue Georges Pompidou
92593 Levallois-Perret Cedex
France

Phone: +33 (0) 1 49 64 52 09
bcs-europe@basf.com

North America

BASF Corporation
North American Regional Headquarters
100 Park Avenue
Florham Park, NJ 07932
USA

Phone: +1-800-962-7831
CosmeticsCustomerCare@basf.com

South America

BASF S.A.
Av. das Nações Unidas, 14171
Crystal Tower
04794-000 - São Paulo - SP
Brazil

Phone: +55 11 2039 2273
personal-care-sa@basf.com

Asia Pacific

BASF East Asia
Personal Care
45/F Jardine House
No.1 Connaught Place
Central, Hong Kong

Phone: +852 2731 0190
personal-care-hk@basf.com

Publisher
BASF Beauty Care Solutions France SAS
3, Rue de Seichamps - 54425 Pulnoy (France)
Email: bcs-europe@basf.com
www.carecreations.basf.com

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